

● RAG Learning

P4 — Explaining Motion

OCR GCSE Physics B (J259) — 21st Century Science · Paper 2 (Depth in Physics)

P4.1 Scalars and Vectors

A **scalar** quantity has magnitude (size) only — e.g. distance, speed, mass, energy, time.

A **vector** quantity has magnitude *and* direction — e.g. displacement, velocity, acceleration, force, momentum, weight.

Scalar	Vector
Distance — total path length	Displacement — straight-line distance <i>and</i> direction from start
Speed — how fast	Velocity — speed in a stated direction

P4.2 Speed, Velocity and Acceleration

Speed: $v = \text{distance} / \text{time}$ (m/s)

Acceleration: $a = \Delta v / t = (v - u) / t$ (m/s²)

u = initial velocity, v = final velocity

SUVAT: $v^2 = u^2 + 2 a s$

$(\text{m/s})^2 = (\text{m/s})^2 + 2 (\text{m/s}^2)(\text{m})$

Typical Speeds

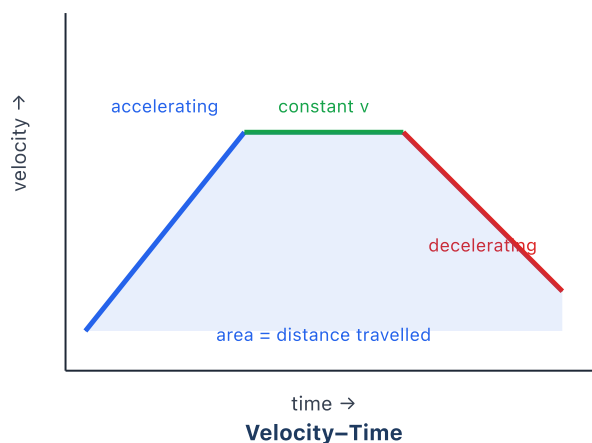
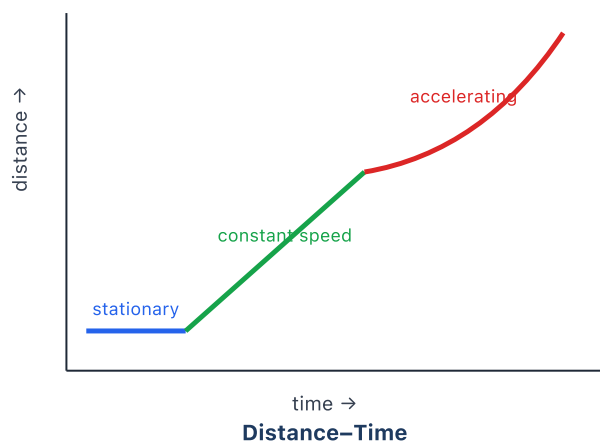
Situation	Typical speed
Walking	1.5 m/s
Running	3 m/s
Cycling	6 m/s
Car (urban)	13 m/s (30 mph)
Car (motorway)	31 m/s (70 mph)
Train	55 m/s
Cruising aeroplane	250 m/s
Sound in air (20 °C)	340 m/s

Worked Example: A car accelerates from 10 m/s to 25 m/s in 5 s. Find its acceleration.

$$a = (v - u)/t = (25 - 10)/5 = \mathbf{3 \text{ m/s}^2}$$

P4.3 Distance–Time and Velocity–Time Graphs

Distance–Time and Velocity–Time Graphs



- On a **distance–time graph**, the *gradient* gives the **speed**.
- On a **velocity–time graph**, the *gradient* gives the **acceleration** and the *area under the line* gives the **distance travelled**.

P4.4 Forces

A **force** is a push or a pull. Forces are vectors, measured in **newtons (N)**. Forces can change an object's shape, speed or direction.

Contact and Non-Contact Forces

- **Contact**: friction, tension, normal reaction, air resistance, drag, upthrust.
- **Non-contact**: gravity, magnetic force, electrostatic force.

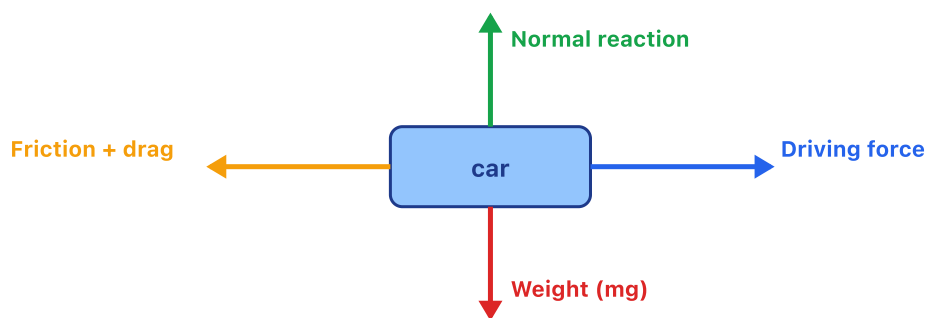
Weight and Mass

$$W = m \times g \quad (\text{N}) = (\text{kg}) \times (\text{N/kg})$$

On Earth, $g \approx 9.8 \text{ N/kg}$. Mass (kg) is the amount of matter; *weight* (N) is the gravitational force acting on that mass.

Free-Body Diagrams

Free-Body Diagram — Car at Constant Speed



At constant velocity, forces balance: driving force = friction + drag.

P4.5 Newton's Laws of Motion

Newton's 1st Law: an object at rest stays at rest, and an object moving in a straight line continues at constant velocity, *unless* acted on by a resultant (net) force.

Newton's 2nd Law: the resultant force on an object equals its mass $\times$ its acceleration.

$$F = m \times a \quad (\text{N}) = (\text{kg}) \times (\text{m/s}^2)$$

Newton's 3rd Law: when object A exerts a force on object B, object B exerts an equal and opposite force on A. These form a *pair* — same size, opposite direction, acting on two different bodies.

Worked Example: A 1200 kg car has a resultant force of 3000 N. What is its acceleration?

$$a = F / m = 3000 / 1200 = \mathbf{2.5 \text{ m/s}^2}$$

P4.6 Inertia and Inertial Mass

Inertia is the tendency of an object to keep doing what it is already doing (staying at rest, or continuing in a straight line at constant velocity). A more massive object has more inertia — it is harder to accelerate.

The ratio **F / a** in Newton's 2nd Law is called the **inertial mass**.

P4.7 Momentum

Momentum (p) = mass $\times$ velocity. Unit: **kg m/s**. Momentum is a *vector*.

$$p = m \times v$$

Momentum is always **conserved** in a closed system (where no external forces act): *total momentum before = total momentum after*.

Worked Example: A 0.5 kg ball moving at 4 m/s hits a stationary 1.5 kg ball and they stick together. Find their common velocity.

Before: $p = 0.5 \times 4 + 1.5 \times 0 = 2 \text{ kg m/s}$

After: $p = (0.5 + 1.5) v = 2v$. So $v = 1 \text{ m/s}$.

Force as Rate of Change of Momentum

$$F = \Delta p / \Delta t = m(v - u) / t$$

A smaller force is needed if the change of momentum happens over a **longer time**. This is why airbags, crumple zones, cycle helmets, crash mats, and seat belts all *increase stopping time* — reducing the peak force on the body during a collision.

P4.8 Stopping Distance

Stopping distance = thinking distance + braking distance.

Thinking distance: distance travelled during the driver's reaction time.

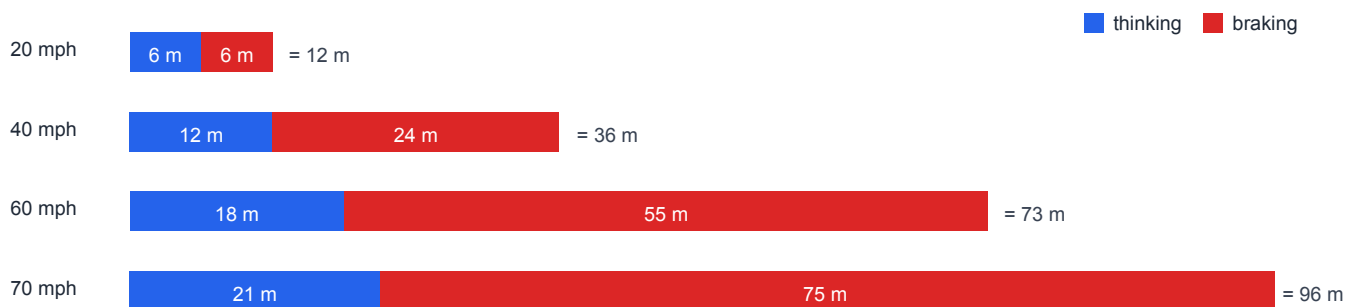
Braking distance: distance travelled while the brakes decelerate the car to rest.

Affects thinking distance	Affects braking distance
Speed	Speed
Tiredness	Road conditions (wet/icy/gravel)
Alcohol/drugs	Tyre condition (tread, pressure)

Distractions (phone)

Brake condition

Stopping Distance at Different Speeds (Highway Code)



Braking distance grows with v^2 — doubling speed roughly quadruples braking distance.

Energy Dissipated in Braking

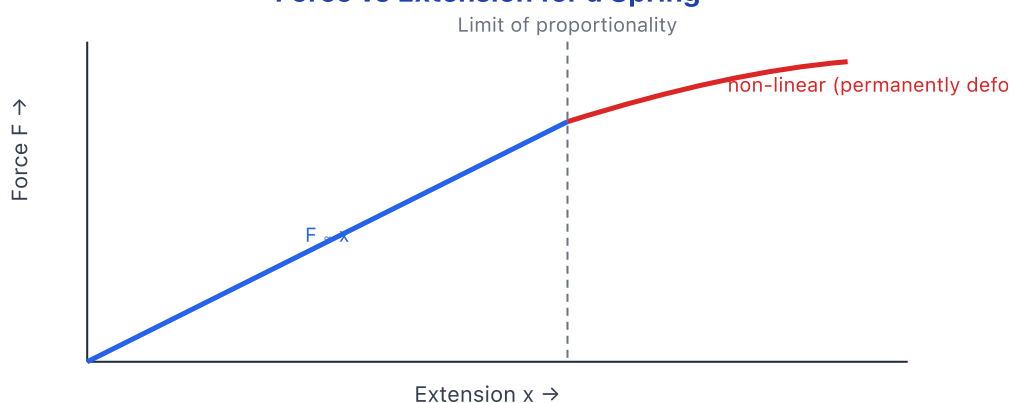
During braking, the car's kinetic energy is transferred to **thermal energy** in the brake discs, pads and tyres (via friction). Because $E_k = \frac{1}{2} m v^2$, doubling the speed means *four times* as much energy to dissipate, so the braking distance also roughly quadruples.

P4.9 Hooke's Law (Forces and Elasticity)

The **extension** of an elastic object (spring, rubber band, wire) is directly proportional to the force applied, provided the limit of proportionality is not exceeded.

$$F = k \times x \quad (\text{N}) = (\text{N/m}) \times (\text{m})$$

Force vs Extension for a Spring



Worked Example: A spring of constant 40 N/m is stretched by 0.15 m. Find the elastic PE stored.

$$E_e = \frac{1}{2} k x^2 = \frac{1}{2} \times 40 \times 0.15^2 = \frac{1}{2} \times 40 \times 0.0225 = \mathbf{0.45 \text{ J}}$$

P4.10 Moments (Turning Forces)

A **moment** is the turning effect of a force about a pivot.

Moment (N m) = Force (N) × perpendicular distance from pivot (m)

Exam Tip: Principle of moments (for a body in equilibrium): total clockwise moments = total anticlockwise moments.

Worked Example: A see-saw is 6 m long, with the pivot in the middle. A 50 kg child sits 1.5 m left of the pivot. Where must a 30 kg child sit to balance it?

$$\text{Left moment} = 50 \times 9.8 \times 1.5 = 735 \text{ N m}$$

$$\text{Right moment} = 30 \times 9.8 \times d = 735 \rightarrow d = 735 / (30 \times 9.8) = \mathbf{2.5 \text{ m}} \text{ to the right.}$$